

ANNs & Deep Learning

Project 3: Plant diseases

Project context

Image recognition is one of the most important use cases in intelligent systems as it allows machines to perceive their environment. In recent years, machine learning approaches have shown to be superior compared to classic computer vision approaches in many use cases. Recently, the domain of smart agriculture has picked up this trend as well. One use case within this area is the camera-based detection of plant diseases, which is a basis for many different high-level applications such as e.g. smart spraying.

Task

Using the “Plant disease recognition dataset”, explore how to create a performant, yet efficient Deep Learning model that classifies the images according to diseases in this dataset.

Explore different approaches/architectures and select an appropriate metric to ensure the quality of the final model.



Source of picture: Rashik Raman, kaggle.com.

Links

Dataset:

- <https://www.kaggle.com/datasets/rashikrahmanpritom/plant-disease-recognition-dataset>